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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/525,290	11/12/2021	Yu Fang	SONA-0006-U01-C06	3029

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GTC Law Group PC & Affiliates
One University Ave., Ste. 302B
Westwood, MA 02090

EXAMINER	
MANIWANG, JOSEPH R	

ART UNIT	PAPER NUMBER
2441	

NOTIFICATION DATE	DELIVERY MODE
02/23/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 17/525,290	Applicant(s) Fang et al.	
	Examiner JOSEPH R MANIWANG	Art Unit 2441	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 11/12/2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-12 and 16-17 is/are rejected.

8) ☑ Claim(s) 2-20 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 11/12/2021 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 11/12/2021.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

2. The information disclosure statement (IDS) submitted on 11/12/2021 was in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the Examiner.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject

to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

3. Claims 1-12, 16, and 17 rejected on the ground of nonstatutory double patenting as being unpatentable over claims 9-20 of U.S. Patent No. 11,349,717. Although the claims at issue are not identical, they are not patentably distinct from each other because the scope of the instant claims is anticipated by the patented claims. For example, compare instant claim 1 with patented claim 9:

U.S. Pat. 11,349,717	App. No. 17/525,290
9. A system, comprising:	1. A system, comprising:
a first network zone of a vehicle comprising a first interconnected plurality of end points; a second network zone of the vehicle comprising a second interconnected plurality of end points;	a first network zone of a mobile system comprising a first interconnected plurality of end points; a second network zone of the mobile system comprising a second interconnected plurality of end points;
a converged network device (CND) interposed between the first network zone and the second network zone, wherein the CND is configured to regulate communications between	a converged network device (CND) interposed between the first network zone and the second network zone, wherein the CND is configured to regulate communications between

end points of the first network zone and the second network zone;	end points of the first network zone and the second network zone;
a first vehicle controller on the first network zone; a second vehicle controller on the second network zone; and	a first mobile system controller on the first network zone; a second mobile system controller on the second network zone; and
a network redundancy circuit structured to selectively provide a regulation control command;	a network redundancy circuit structured to selectively provide a regulation control command;
wherein: the CND is further configured to adjust the regulating the communications between end points of the first network zone and end points of the second network zone in response to the regulation control command; the regulation control command comprises an off-nominal condition corresponding to the first vehicle controller; the off-nominal condition comprises a loss of a data element, wherein the adjusted regulating the communications comprises providing an alternate data element to the first vehicle controller; and the lost data element includes a first sensor and the alternate data	wherein the CND is further configured to adjust the regulating the communications between end points of the first network zone and end points of the second network zone in response to the regulation control command.

element includes a second sensor distinct from the first sensor.	

Instant dependent claims 2-12, 16, and 17 are similarly anticipated by patented claims 9-20. Accordingly, instant claims 1-12, 16, and 17 are rejected.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

4. Claim 1 is rejected under 35 U.S.C. 103 as being unpatentable over Sawal et al. (U.S. Pat. App. Pub. 2014/0359185), hereinafter Sawal, and further in view of Nagapudi et al. (U.S. Pat. App. Pub. 2011/0078299), hereinafter Nagapudi.

Regarding claim 1, Sawal disclosed a system, comprising:

a first network zone of a mobile system (local area network, ¶[0020]) comprising a first interconnected plurality of end points (local area network coupled, i.e., interconnected, to a plurality of additional devices, i.e., end points, ¶[0021]);

a second network zone of the mobile system (storage area network, ¶[0020]) comprising a second interconnected plurality of end points (storage area network coupled, i.e., interconnected, to a plurality of storage devices, i.e., end points, ¶[0021]);

a converged network device (CND) interposed between the first network zone and the second network zone (converged network device configured to receive communications from more than one type of network, i.e., interposed between the first network zone and the second network zone, ¶[0020], Fig. 1, element 102 connected to elements 110 and 122), wherein the CND is configured to regulate

communications between end points of the first network zone and the second network zone (converged network device configured to facilitate, i.e., regulate, the retrieval and use of information stored in the storage area network by a host in the local area network, i.e., communications between end points of the first network zone and the second network zone, ¶[0022]);

a first mobile system controller on the first network zone (host, i.e., first mobile system controller, in local area network, ¶[0021]);

a second mobile system controller on the second network zone (storage device, i.e., second mobile system controller, in storage area network, ¶[0021]).

Sawal did not disclose:

a network redundancy circuit structured to selectively provide a regulation control command;

wherein the CND is further configured to adjust the regulating the communications between end points of the first network zone and end points of the second network zone in response to the regulation control command.

Nagapudi disclosed:

a network redundancy circuit structured to selectively provide a regulation control command (baseboard management controller/BMC, i.e., network redundancy circuit, providing a configuration command, i.e., regulation control command, ¶[0044]);

wherein the CND (adapter/converged network adapter/CNA, i.e., CND, ¶[0007], Fig. 1 and 2) is further configured to adjust the regulating the communications between end points of the first network zone and end points of the second network zone in response to the regulation control command (reconfiguring, i.e., adjusting, the adapter according to, i.e., in response to, the configuration command, ¶[0046]-[0047]; adapter interfacing, i.e., regulating, connection, i.e., communications, between CEE networks and FC SANs, i.e., end points of the first network zone and end points of the second network zone, ¶[0007]).

It would have been obvious to one of ordinary skill in the art before the effective filing date of the invention to modify the system of Sawal to include a network redundancy circuit structured to selectively provide a regulation control command, wherein the CND is further configured to adjust the regulating the communications between end points of the first network zone and end points of the second network zone

in response to the regulation control command as claimed, because doing so would have been a simple way to reconfigure adapters as desired without extra steps, e.g., BIOS updates (Nagapudi, ¶[0046]), which can be cumbersome and disruptive (Nagapudi, ¶[0009]). Additionally, doing so would have been use of known technique (i.e., adjusting CND behavior/regulating via command from a circuit) to improve similar devices/methods/products (i.e. the CND of Sawal) in the same way.

Allowable Subject Matter

5. Claims 2-12, 16, and 17 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims, and upon the filing of a proper Terminal Disclaimer to overcome the above double patenting rejection.

6. Claims 13-15 and 18-20 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JOSEPH R MANIWANG whose telephone number is (571)270-7257. The examiner can normally be reached 8:30AM - 4:30PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wing F Chan can be reached on (571) 272-7493. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JOSEPH R MANIWANG/
Examiner, Art Unit 2441